CSE 1000. FRESHMAN UNDERGRADUATE RESEARCH. 0 Hours.

Freshman level undergraduate research course. Prerequisites: Departmental good standing and permission of instructor. May be taken a maximum of 3 times.

CSE 1105. INTRODUCTION TO COMPUTER SCIENCE AND ENGINEERING. 1 Hour.

Introduction to engineering concepts, the computer science and engineering disciplines, skills for written communication, and departmental orientation.

CSE 1106. INTRODUCTION TO COMPUTER SCIENCE AND ENGINEERING. 1 Hour.

A practical approach to hands-on computer hardware and software systems in a laboratory environment. Students will be exposed to basic engineering concepts such as simple circuits, digital logic, embedded controllers, computer networking, software design, and Linux operating systems. Prerequisite: C or better in CSE 1310.

CSE 1205. INTRODUCTION TO COMPUTER SCIENCE AND ENGINEERING. 2 Hours.

A practical approach to hands-on computer hardware and software systems in a laboratory environment. Students will be exposed to basic engineering concepts such as simple circuits, digital logic, embedded controllers, computer networking, software design, and Linux operating systems. Some College of Engineering requirements are satisfied by the content of this course. Prerequisite: CSE 1310.

CSE 1301. COMPUTER LITERACY. 3 Hours. (TCCN = COSC 1301)

For those persons having an interest in finding out what a computer is (and is not), the types of problems suited for computers, and how to utilize a computer to solve problems. The organization and characteristics of computers; application of commercial software such as word processors, spreadsheets, database packages, and communications packages.

CSE 1310. INTRODUCTION TO COMPUTERS and PROGRAMMING. 3 Hours. (TCCN = COSC 1320)

An introduction to the computer, to the algorithmic process, and to programming using basic control and data structures, using a procedural language. Prerequisite: C or better in MATH 1302 or MATH 1402 or C or better in (or concurrent enrollment in) a subsequent mathematics course (MATH 1421, MATH 1426, MATH 2425, MATH 2326, MATH 3330, HONR-SC 1426 or HONR-SC 2425) or a qualifying score on the ALEKS PPL and C or better in UNIV 1131 (or concurrent enrollment) or ENGR 1101 (or concurrent enrollment).

CSE 1311. INTRODUCTION TO PROGRAMMING FOR ENGINEERS. 3 Hours.

An introduction to the computer, to the algorithmic process, and to programming using basic control and data structures. This class is currently using the C language. Prerequisite: C or better in (or concurrent enrollment in) one of the following; (MATH 1421, MATH 1426, MATH 2425, MATH 2326, MATH 3330, HONR-SC 1426, or HONR-SC 2425).

CSE 1320. INTERMEDIATE PROGRAMMING. 3 Hours.

Programming concepts beyond basic control and data structures. Emphasis is given to data structures including linked-lists and trees as well as modular design consistent with software engineering principles. Prerequisite: C or better in CSE 1310 and C or better in (or concurrent

enrollment in) (MATH 1421, MATH 1426, MATH 2425, MATH 2326, MATH 3330, HONR-SC 1426, or HONR-SC 2425) or a qualifying score on the ALEKS PPL and C or better in UNIV 1131 (or concurrent enrollment) or ENGR 1101 (or concurrent enrollment.).

CSE 1325. OBJECT-ORIENTED PROGRAMMING. 3 Hours.

Object-oriented concepts, class diagrams, collection classes, generics, polymorphism, and reusability. Projects involve extensive programming and include graphical user interfaces and multithreading. Prerequisite: CSE 1320.

CSE 1326. OBJECT-ORIENTED PROGRAMMING IN C++. 3 Hours.

Object-Oriented concepts (abstraction, encapsulation, inheritance, polymorphism) in the context of programming in C++ 11 and later. Projects involve extensive programming in C++. Prerequisite: CSE 1320.

CSE 1392. SPECIAL TOPICS. 3 Hours.

New developments in the field of computer science and engineering. Topic may vary from semester to semester. May be repeated for credit when topic changes. Departmental approval required in advance to use for degree credit. Prerequisite: consent of advisor.

CSE 2000. SOPHOMORE UNDERGRADUATE RESEARCH. 0 Hours.

Sophomore level undergraduate research course. Prerequisites: Departmental good standing and permission of instructor. May be taken a maximum of 3 times.

CSE 2100. PRACTICAL COMPUTER HARDWARE/SOFTWARE SYSTEMS. 1 Hour.

A practical approach to hands-on computer hardware and software systems in a laboratory environment. Students will be exposed to basic design concepts using off-the-shelf hardware components and to tools that enable the design of complex software systems. Prerequisite: CSE 1320.

CSE 2312. COMPUTER ORGANIZATION and ASSEMBLY LANGUAGE PROGRAMMING. 3 Hours.

Computer organization from the viewpoint of software, including instruction set architectures, memory addressing, integer and floating-point representation and arithmetic, instruction pipelining, cache, memory virtualization, and I/O. The relationship of higher-level programming languages to assembly language and instruction set architecture is also explored. Prerequisite: a C or better in CSE 1320 and a C or better in CSE 1205 or CSE 1106.

CSE 2315. DISCRETE STRUCTURES. 3 Hours.

Propositional and predicate logic, mathematical proof techniques, sets, combinatorics, functions and relations, graphs, and graph algorithms. Prerequisite: C or better in CSE 1310 and C or better in MATH 1426 (or C or better in or concurrent enrollment in MATH 2425).

CSE 2392. SPECIAL TOPICS. 3 Hours.

New developments in the field of computer science and engineering. Topic may vary from semester to semester. May be repeated for credit when topic changes. Departmental approval required in advance to use for degree credit. Prerequisite: consent of advisor.

CSE 2440. CIRCUIT ANALYSIS. 4 Hours.

Basic principles of electrical circuits using resistors, capacitors and inductors. Filter analysis and synthesis using complex algebra. Introduction to operational amplifiers. Time domain and frequency domain analysis and taxonomy of signals. Concurrent laboratory experiments complement lecture topics. Prerequisite: Grade C or better in MATH 2425 and PHYS 1444.

CSE 2441. DIGITAL LOGIC DESIGN I. 4 Hours.

Analysis, design and testing of combinational and sequential logic circuits. Topics include Boolean algebra, logic circuit minimization techniques, synchronous sequential circuit design, algorithmic state machine design, design of arithmetic/logic and control units, and Verilog programming of FPGA devices. Prerequisite: CSE 1320 and CSE 2315.

CSE 3000. JUNIOR UNDERGRADUATE RESEARCH. 0 Hours.

Junior level undergraduate research course. Prerequisites: Departmental good standing and permission of instructor. May be taken a maximum of 3 times.

CSE 3302. PROGRAMMING LANGUAGES. 3 Hours.

Introduction, analysis, and evaluation of the important concepts found in a variety of programming languages. Formalisms useful in specifying language syntax and semantics; programming language paradigms such as algorithmic, functional, logic, and object-oriented. Prerequisite: C or better in each of the following: CSE 1325, CSE 2312 and CSE 3318.

CSE 3310. FUNDAMENTALS OF SOFTWARE ENGINEERING. 3 Hours.

Software engineering principles, processes, and techniques; software development approaches focusing on functional analysis and functional design methods. Configuration management, implementation strategies, and testing. Team project. Prerequisite: C or better in each of the following: CSE 1320, CSE 1325 and CSE 2315.

CSE 3311. OBJECT-ORIENTED SOFTWARE ENGINEERING. 3 Hours.

Study of an agile unified methodology and its application to object-oriented software development. Topics include requirements acquisition, use case derivation, modeling and design of interaction behavior and state behavior, introduction to design patterns, derivation of design class diagrams, implementation considerations and deployment. Team project. Prerequisite: C or better in each of the following: CSE 3310 and CSE 3318.

CSE 3313. INTRODUCTION TO SIGNAL PROCESSING. 3 Hours.

Examines models for presentation and processing of digital signals. Sampling theorem, correlation and convolution, time and frequency analysis of linear systems, Fourier transform, Z-transform, design of digital filters structures for discrete time systems. Prerequisites: C or better in each of the following: CSE 3318, CSE 3380 or MATH 3330, and IE 3301 or MATH 3313.

CSE 3314. PROFESSIONAL PRACTICES. 3 Hours.

Ethics. Contemporary social aspects and responsibilities of computing in a global, societal context. Lifelong learning goals and resources. Entrepreneurship and intellectual property. Project involving written and oral communication. Prerequisite: C or better in CSE 3318 and COMS 2302.

CSE 3315. THEORETICAL CONCEPTS IN COMPUTER SCIENCE AND ENGINEERING. 3 Hours.

Selected theoretical concepts including regular and context free languages, finite state and pushdown automata, Turing machines, computability, and NP-completeness. Prerequisite: C or better in CSE 2315.

CSE 3318. ALGORITHMS and DATA STRUCTURES. 3 Hours.

Design and analysis of algorithms with an emphasis on data structures. Approaches to analyzing lower bounds on problems and upper bounds on algorithms. Classical algorithm design techniques including algorithms for sorting, searching, and other operations on data structures such as hash tables, trees, graphs, strings, and advanced data structures, dynamic programming and greedy approaches. Prerequisite: CSE 1320 and CSE 2315.

CSE 3320. OPERATING SYSTEMS. 3 Hours.

Functions and components of an operating system, including process synchronization, job scheduling, memory management, file systems protection, and deadlocks. Related system software, such as loaders, linkers, assemblers, and windowing systems. Prerequisite: C or better in CSE 2312.

CSE 3323. ELECTRONICS. 3 Hours.

Design, analysis and testing of electronic circuits. Topics include operational amplifiers, diodes, bipolar-junction transistors (BJTs), and field-effect transistors (FETs) and their applications. Concurrent laboratory experiments complement lecture topics. Prerequisite: C or better in CSE 2440.

CSE 3330. DATABASE SYSTEMS AND FILE STRUCTURES. 3 Hours.

Database system architecture; file structures for databases, including indexing hashing, and B+-trees; the relational model and algebra; the SQL database language; Entity-Relationship data modeling; functional dependencies and basic normalization. Prerequisite: C or better in each of the following: CSE 1325 and CSE 3318.

CSE 3340. INTRODUCTION TO HUMAN COMPUTER INTERACTION. 3 Hours.

Introduction to the interdisciplinary field of Human-Computer Interaction. Studio-based course that bridges cognitive science, ethnography, and anthropology to provide a profound understand of user experience, while concurrently fostering frontend development skills in software, hardware, and wireless communication for the crafting of state-of-the-art user interfaces. Emphasizing a human-centered design ethos, the course encourages the design of interfaces that are not only functional but also meaningful and critically reflective. Prerequisite: C or better in CSE 3318, and C or better in CSE 3310.

CSE 3341. DIGITAL LOGIC DESIGN II. 3 Hours.

Hierarchical organization, design, simulation, implementation, and testing of digital systems. Industrial standard computer-aided design tools including hardware description languages (HDLs), field-programmable gate arrays (FPGAs), and other prototyping hardware and software will be employed. Design of arithmetic and other algorithmic processes will be covered. A term project will be required. Prerequisite: C or better in CSE 2441.

CSE 3380. LINEAR ALGEBRA FOR CSE. 3 Hours.

Solving systems of equations, matrix algebra, determinants, vector spaces, orthogonality and least squares, with applications to computer science. Prerequisite: C or better in CSE 2315.

CSE 3392. SPECIAL TOPICS. 3 Hours.

New developments in the field of computer science and engineering. Topic may vary from semester to semester. May be repeated for credit when topic changes. Departmental approval required in advance to use for degree credit. Prerequisite: consent of advisor.

CSE 3442. EMBEDDED SYSTEMS I. 4 Hours.

Design of microcontroller-based systems, including microprocessor programming, component and system architectures, memory interfacing, asynchronous and synchronous serial interfaces, timer-based peripherals, analog to digital (A/D) and digital to analog (D/A) converters, and typical applications. Prerequisite: C or better in each of the following: CSE 2312, CSE 2440 and CSE 2441.

CSE 4000. SENIOR UNDERGRADUATE RESEARCH. 0 Hours.

Senior level undergraduate research course. Prerequisites: Departmental good standing and permission of instructor. May be taken a maximum of 3 times.

CSE 4191. INDIVIDUAL PROJECTS. 1 Hour.

Special problems in computer science and engineering on an individual basis. Topics may change from semester to semester. May be repeated for credit. Departmental approval must be obtained in advance for degree credit. Prerequisite: consent of instructor and department chairperson.

CSE 4303. COMPUTER GRAPHICS. 3 Hours.

Theory and practice for the visual representation of data by computers including display devices, output primitives, planes and curved surfaces, two- and three-dimensional transformations, parallel and perspective viewing, removal of hidden lines and surfaces, illumination models, ray tracing, radiosity, color models, and computer animation. Prerequisite: Admitted into an Engineering Professional Program. C or better in each of the following: CSE 3318, and either CSE 3380 or MATH 3330.

CSE 4304. GAME DESIGN AND DEVELOPMENT. 3 Hours.

Fundamentals of what it takes to create a game that is simultaneously economically attractive enough to produce as well as being fun to play. The former characteristic is objective, though often tricky to bound adequately while the latter is subjective, though easily demonstrated. This is a team-based class; each team conceives, designs, and develops a complete game through a series of exercises that build upon each other and using freely available tools. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3380 or MATH 3330.

CSE 4305. COMPILERS FOR ALGORITHMIC LANGUAGES. 3 Hours.

Review of programming language structures, translation, and storage allocation. Theory and practice of compilers and issues in compiler construction including parsing, intermediate code generation, local optimization problems such as register allocation, data-flow analysis, and

global optimization. Prerequisite: Admitted into an Engineering Professional Program. C or better in the following: CSE 3302 and CSE 3315.

CSE 4308. ARTIFICIAL INTELLIGENCE. 3 Hours.

An introduction to the field of artificial intelligence studying basic techniques such as heuristic search, deduction, learning, problem solving, knowledge representation, uncertainty reasoning and symbolic programming languages such as LISP. Application areas may include intelligent agents, data mining, natural language, machine vision, planning and expert systems. Prerequisite: Admitted into an Engineering Professional Program. C or better in each of the following: CSE 3318 and (IE 3301 or MATH 3313).

CSE 4309. FUNDAMENTALS OF MACHINE LEARNING. 3 Hours.

This course offers an introduction to machine learning. Topics include naive Bayes classifiers, linear regression, linear classifiers, neural networks and backpropagation, kernel methods, decision trees, feature selection, clustering, and reinforcement learning. A strong programming background is assumed, as well as familiarity with linear algebra (vector and matrix operations), and knowledge of basic probability theory and statistics. Prerequisites: Admitted into an Engineering Professional Program. C or better in each of the following: CSE 3318, IE 3301 or MATH 3313, and CSE 3380 or MATH 3330.

CSE 4310. FUNDAMENTALS OF COMPUTER VISION. 3 Hours.

This course introduces students to basic concepts and techniques in computer vision. The topics covered include morphological operations, connected component analysis, image filters, edge detection, feature extraction, object detection, object recognition, tracking, gesture recognition, image formation and camera models, calibration, and stereo vision. A strong programming background is assumed, as well as familiarity with linear algebra (vector and matrix operations), and knowledge of basic probability theory and statistics. Prerequisite: Admitted into an Engineering Professional Program. C or better in each of the following: CSE 3318, IE 3301 or MATH 3313, and CSE 3380 or MATH 3330.

CSE 4311. NEURAL NETWORKS AND DEEP LEARNING. 3 Hours.

This course offers an introduction to neural networks and deep learning. Topics include perceptrons, single-layer neural networks, multi-layer neural networks, Tensorflow and Keras, convolutional neural networks, transfer learning, deep learning methods for object recognition and object detection in images, and sequential learning models for analyzing text. Auto-encoders and generative adversarial networks will be covered to some extent. A strong programming and algorithmic background is assumed, as well as familiarity with linear algebra (vector and matrix operations). Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3380 or MATH 3330, and C or better in IE 3301 or MATH 3313.

CSE 4314. PROFESSIONAL PRACTICES. 3 Hours.

Ethics. Contemporary social aspects and responsibilities of computing in a global, societal context. Lifelong learning goals and resources. Entrepreneurship and intellectual property. Project involving written and oral communication. Prerequisite: Admitted into an Engineering Professional Program. C or better in COMS 2302.

CSE 4316. COMPUTER SYSTEM DESIGN PROJECT I. 3 Hours.

Analysis and design of an industry-type project that involves hardware and software components to meet desired needs within realistic constraints and standards. The project is to be completed in CSE 4317 the following semester. Multidisciplinary teams of CSE 4316 students are required to develop, review, and present problem definition, project planning, requirements formulation, and design specification. Prerequisites: Admitted into a CSE Professional Program. For academic plan CS or SE, C or better in CSE 3310 and CSE 3320, and C or better in CSE 3314 (or concurrently). For academic plan CSE_CP, C or better in CSE 3320 and CSE 3442, and C or better in CSE 3314 (or concurrently).

CSE 4317. COMPUTER SYSTEM DESIGN PROJECT II. 3 Hours.

Implementation, integration, quality assurance through peer review and testing, and deployment of the project designed in CSE 4316; oral presentation, documentation and project demonstration. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 4316 and continuation with the same team.

CSE 4321. SOFTWARE TESTING and MAINTENANCE. 3 Hours.

Study of software quality assurance, software testing, and software maintenance processes, methods and techniques including formal review techniques, software verification, validation, and testing, types of software maintenance, maintenance activities, and regression testing. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3310.

CSE 4322. SOFTWARE PROJECT MANAGEMENT. 3 Hours.

Introduction to software project management. Issues include effort estimation and costing, project planning and scheduling, option analysis, software quality assurance, and formal technical reviews. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3310.

CSE 4323. QUANTITATIVE COMPUTER ARCHITECTURE. 3 Hours.

Pipelined processors, parallel processors including shared and distributed memory, multicore, Very Long Instruction Word (VLIW) and graphics processors, memory and cache design, computer peripherals, and computer clusters. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3320.

CSE 4331. DATABASE IMPLEMENTATION AND THEORY. 3 Hours.

Review of the relational model and algebra; relational calculus; relational database design theory; advanced data modeling concepts; object-oriented and object-relational databases; database system implementation techniques, including concurrency control, recovery, atomic commitment, and query processing and optimization, database security; introduction to advanced concepts, such as active, deductive, spatial, temporal, multimedia and distributed databases. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3330.

CSE 4333. CLOUD COMPUTING FUNDAMENTALS AND APPLICATIONS. 3 Hours.

Cloud Computing has transformed the IT industry by opening the possibility for infinite or at least highly elastic scalability in the delivery of enterprise applications and software as a service (SaaS). Cloud computing is a method of computing where a shared group of resources such as file storage, web servers, data processing services, and applications are accessed via a public internet connection or a private VPN and/or direct network connection. Cloud computing gives mature

enterprises and new start-ups the option to deploy their applications to systems of infinite computational power with practically no initial capital investment and with modest operating costs proportional to the actual use. Examples of cloud computing services include Amazon Web Services, Microsoft Azure, Google Cloud Platform, and IBM Softlayer. This course introduces students to the fundamental and design concepts of writing software applications for the cloud. This course will discuss tools required by an application programmer in building, deploying, and maintaining cloud applications. A combination of lectures and lab activities will expose students to the programming interface utilized in developing cloud applications. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3330, and C or better in CSE 3320.

CSE 4334. DATA MINING. 3 Hours.

Automatic discovery of patterns and knowledge from large data repositories, including databases, data warehouses, Web, document collections, and transactions. Basic topics of data mining including data preprocessing, data warehousing and online analytical processing (OLAP), data cube, frequent pattern and association rule mining, correlation analysis, classification and prediction and clustering, as well as advanced topics covering the techniques and applications of data mining on Web and text documents. Prerequisite: Admitted into an Engineering Professional Program. C or better in each of the following: IE 3301 (or MATH 3313). Co-requisite: CSE 3330.

CSE 4340. FUNDAMENTALS OF WIRELESS NETWORKS. 3 Hours.

Fundamentals of wireless networks, radio spectrum, coding and modulation, multiple access techniques, antennas, noise and interference, channels, demodulation and decoding, error rates and capacity, link budgets, medium access control, rate adaption, and wireless LAN/PAN, ad-hoc, and sensor networks. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 4344 or CSE 4352 or consent of instructor.

CSE 4342. EMBEDDED SYSTEMS II. 3 Hours.

Advanced course in design of microcontroller-based systems. Emphasis is on the application of microcontrollers to real-time problems. Topics include the study of the differences in bare metal and embedded Linux implementations, simple Linux character device drivers, bootloader design, watchdog and supervision concepts, and developing applications such as PID controllers. Course includes significant laboratory content and a project with extensive hardware and software requirements. Prerequisite: Admitted into an Engineering Professional Program. C or better in each of the following: CSE 3323, CSE 3442, and CSE 3313.

CSE 4344. COMPUTER NETWORK ORGANIZATION. 3 Hours.

Design and analysis of computer networks. Emphasis on the OSI architecture but discusses other schemes (e.g., ARPAnet). Data link control, local networks, protocols/architectures, network access protocols, transport protocols, internetworking, and ISDN. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3320.

CSE 4345. COMPUTATIONAL METHODS. 3 Hours.

Introduction to numerical methods for solving problems in computer science and computer engineering. Topics include computer arithmetic, linear and nonlinear equations, eigenvalue

problems, least squares, optimization, interpolation, and simulation. Prerequisite: Admitted into an Engineering Professional Program. C or better in each of the following IE 3301 or MATH 3313, CSE 3318, and either CSE 3380 or MATH 3330.

CSE 4351. PARALLEL PROCESSING. 3 Hours.

Theory and practice of parallel processing, including characterization of parallel processors, models for memory, algorithms, and interprocess synchronization. Issues in parallelizing serial computations, efficiency and speedup analysis. Programming exercises using one or more concurrent programming languages, on one of more parallel computers. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3320.

CSE 4352. IOT AND NETWORKING. 3 Hours.

Study of protocol stacks and layers, implementation of an Ethernet protocol stack, and design of a basic low-latency, small footprint IoT protocol on bare metal embedded devices and embedded Linux systems. Course includes multiple projects with hardware construction and extensive software and integration requirements. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3442.

CSE 4354. REAL-TIME OPERATING SYSTEMS. 3 Hours.

Implementation of a real-time operating system with cooperative and preemption context switching, priority scheduling, semaphores, message queues, and inter-process communications on bare metal microcontrollers. Course includes multiple projects with hardware construction and rigorous software requirements. Prerequisite: Admitted into an Engineering Professional Program. C or better in both CSE 3320 and CSE 3442.

CSE 4355. ELECTROMECHANICAL SYSTEMS AND SENSORS. 3 Hours.

Applications of electronics and microcontrollers to the control of electromechanical systems. Topics include driving brushless motors (including stepper motors), brushed permanent magnet motors, and other mechanical actuators; the use of the sensors including IMU,LIDAR, RADAR, GPS, capacitive/inductive sensing, laser distance, thermocouples, strain, pressure, optical encoders, and Hall devices; and control applications. Course includes significant laboratory content and a project with extensive hardware and software requirements. Prerequisite: Admitted into an Engineering Professional Program. C or better in both CSE 3323 and CSE 3442.

CSE 4356. SYSTEM ON CHIP (SOC) DESIGN. 3 Hours.

Design of FPGA-based system on chip solutions, including processor subsystems, FPGA fabric, processor to FPGA bridges, and Linux device drivers. Course includes a project with extensive software requirements. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3442.

CSE 4358. MICROPROCESSOR SYSTEMS. 3 Hours.

Asynchronous and synchronous memory interfacing and timing, design and implementation of DMA controllers and SDRAM controllers. Course includes a project with significant system design. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3442.

CSE 4360. AUTONOMOUS ROBOT DESIGN AND PROGRAMMING. 3 Hours.

An introduction to robotics and the design and programming of autonomous robot systems. Topics include basic kinematics, dynamics, and control, as well as sensors, knowledge representation, and programming techniques. Course work includes individual and group projects involving the building and programming of simulated and real robots. Prerequisite: Admitted into an Engineering Professional Program. C or better in each of the following: CSE 3318, CSE 3320 and CSE 3380 (or MATH 3330).

CSE 4361. SOFTWARE DESIGN PATTERNS. 3 Hours.

In-depth study of software design patterns including description of patterns, design principles and techniques used by patterns as well as application of patterns to solving practical design problems. Team project. Prerequisites: Admitted into an Engineering Professional Program. C or better in CSE 3311.

CSE 4362. GAME ANALYSIS. 3 Hours.

Examines the technical, formal, and cultural elements of games. Teaches game design concepts through hands-on analysis and creative practice. Prerequisite: Admitted into an Engineering Professional Program.

CSE 4363. ESPORTS. 3 Hours.

Surveys the esports industry from the perspective of multiple professions, including coaching, design, business, entertainment, and cultural production. Combines critical study with experiential learning through activities such as live analysis, design exercises, and industry engagement. Students identify a concrete need in an esports context and design a computational or interactive system to address it, drawing on principles of training, simulation, and performance analysis. Prerequisite: Admitted into an Engineering Professional Program.

CSE 4372. RISC PROCESSOR DESIGN. 3 Hours.

Design of a RISC processor, based on RISC V and custom instruction set architectures with implementation on an FPGA target for test and verification. Course includes a project with extensive software requirements. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3442.

CSE 4373. GENERAL PURPOSE GPU PROGRAMMING. 3 Hours.

Study of general purpose computation on a GPU. Topics include GPU architectures, stream processing, and programming languages such as OpenCL and CUDA that realize data-parallel, high-throughput compute kernels on GPU architectures. Prerequisites: Admitted into an Engineering Professional Program. C or better in CSE 3320.

CSE 4376. DIGITAL COMMUNICATION SYSTEMS. 3 Hours.

Study of digital communication systems including source and channel coding, digital modulation techniques, inter-symbol interference, and multi-channel combining and multiple-access methods. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3313.

CSE 4377. WIRELESS COMMUNICATION SYSTEMS. 3 Hours.

Study of wireless systems including modulation, amplification, linearization techniques, filtering, antennas, propagation, reception, and demodulation. Topics include software defined radio design, link budget, and interference analysis. Course includes significant laboratory content.

Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3313 and CSE 3442.

CSE 4378. INTRODUCTION TO UNMANNED VEHICLE SYSTEMS. 3 Hours.

Introduction to UVS (Unmanned Vehicle Systems) such as UAS (Unmanned Aircraft Systems), UGS (Unmanned Ground System) and UMS (Unmanned Maritime System), their history, missions, capabilities, types, configurations, subsystems, and the disciplines needed for UVS development and operation. UVS missions could include student competitions sponsored by various technical organizations. This course is team-taught by engineering faculty. Prerequisite: Admission to a professional engineering or science program.

CSE 4379. UNMANNED VEHICLE SYSTEM DEVELOPMENT. 3 Hours.

Introduction to the technologies needed to create an UVS (Unmanned Vehicle System). Integration of these technologies (embodied as a set of sensors, actuators, computing and mobility platform sub-systems) into a functioning UVS through team work. UVS could be designed to compete in a student competition sponsored by various technical organizations or to support a specific mission or function defined by the instructors. This course is team-taught by engineering faculty. Prerequisite: B or better in CSE 4378 and admission to the UVS certificate program.

CSE 4380. INFORMATION SECURITY. 3 Hours.

Hands-on introduction to the basics of security. Includes system security, buffer overflows, a high-level overview of cryptography, firewalls and intrusion detection/prevention, malware, penetration testing, forensics, and system administration. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3320.

CSE 4381. INFORMATION SECURITY II. 3 Hours.

Deeper study of the fundamentals of security, including symmetric key cryptography, public key cryptography, cryptographic protocols, malware design, network attacks and defenses, data security, privacy, and wireless security. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3320 and C or better in CSE 4344 (or concurrently).

CSE 4382. SECURE PROGRAMMING. 3 Hours.

This course is an introduction to methods of secure software design and development. Students will learn about the major security problems found in software today. Using this knowledge, they will work in teams to find these bugs in software, fix the bugs, and design software so that it has fewer security problems. Static analysis tools will be a core part of the class, but students will also be exposed to black box testing tools. Topics will include input validation, buffer overflow prevention, error handling, web application issues, and XML. Prerequisite: Admitted into an Engineering Professional Program. C or better in CSE 3320.

CSE 4391. INDIVIDUAL PROJECTS. 3 Hours.

Special problems in computer science and engineering on an individual basis. Topics may change from semester to semester. May be repeated for credit. Departmental approval must be obtained in advance for degree credit. Prerequisite: consent of instructor and department chairperson.

CSE 4392. SPECIAL TOPICS. 3 Hours.

New developments in the field of computer science and engineering. Topic may vary from semester to semester. May be repeated for credit when topic changes. Departmental approval required in advance to use for degree credit. Prerequisite: consent of instructor.